

Hossam Eid

AI/ML Engineer · Data Scientist

Kazan, Russia · open to relocation (Moscow, Saint Petersburg, Novosibirsk) · remote, hybrid, on-site

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Permanent residence permit (RF), unrestricted work authorisation

Full-cycle ML/AI engineer: data preparation → model training and validation → REST API → Docker deployment.

Primary focus — LLM/RAG and multi-agent systems. Analytical foundation: applied statistics and hypothesis testing,

A/B design, SQL with window functions, EDA, model interpretation with SHAP. Additionally — time series and

predictive analytics, computer vision.

Experience

Independent AI/ML projects (open source) — ML/AI Engineer

01.2024 — present

Kazan · github.com/hossameid7

LLM, RAG and multi-agent systems, 2026

- Multi-Agent Log Analyzer: designed an autonomous three-agent system on LangGraph for IT-infrastructure monitoring. RAG pipeline on ChromaDB, report validation through Pydantic v2, support for local Ollama and OpenAI-compatible APIs. Covered by 27+ pytest tests.
- LLM Guardrails Safety Gateway: built an asynchronous security gateway on FastAPI — PII masking, prompt injection and jailbreak blocking (35+ rules). A ChromaDB semantic cache returns answers in under 10 ms. Set up p50/p95/p99 latency telemetry.
- NLP assistant for legal PDF contracts: segmentation into semantic blocks, context-aware translation, automatic glossary extraction.

Stack: Python, LangGraph, LangChain, ChromaDB, FastAPI, Hugging Face, Pydantic v2, Docker.

Remaining useful life prediction for industrial equipment, 2025

The work passed technical review by TVEL (Rosatom).

- Ran exploratory analysis on a 3,000-record dataset (H₂, CO, C₂H₄, C₂H₂): distributions, correlations, missing values, outliers.
- Built time-series feature engineering (tsfresh, TSFEL), performed feature selection and attribution with SHAP.
- Classification (4 imbalanced classes): stacking of Random Forest, SVM and XGBoost with a Logistic Regression meta-model — accuracy 0.97, macro-F1 0.92 (macro-F1 chosen because of the imbalance).
- Regression: compared 11 models (LightGBM, CatBoost, GRU/LSTM) on a single validation scheme — best LightGBM, R² 0.86, MAE 66.3, RMSE 91.83.
- Designed a microservice architecture: Django REST Framework + a dedicated Flask ML-serving API, JWT, WebSockets; React + TypeScript dashboard with Plotly.js charts. Chat assistant on a local Qwen 2.5.

Stack: Python, pandas, NumPy, scikit-learn, XGBoost, LightGBM, CatBoost, tsfresh, SHAP, Django REST Framework, Flask, Docker.

Computer vision, medical imaging, 2026 (research prototypes, not medical devices)

- Brain tumour detection from MRI: Xception — test accuracy 98.02%; U-Net with a VGG16 backbone — val. accuracy 98.46%. Implemented on-device inference through native C++ modules.
- Tumour segmentation: U-Net with batch normalisation — val. accuracy 94.7%; Streamlit demo.

Stack: Python, TensorFlow, Keras, OpenCV, C++, Flutter.

Other projects: Consent Manager (Russian data-protection law 152-FZ, Flask + SQLAlchemy); a Telegram AI-assistant builder on LLM + RAG.

Kazan Federal University (KFU), ITIS — ML/AI technical consultant (bachelor theses)

09.2024 — present

Kazan

I support my academic supervisor with the technical side of bachelor ML/AI theses. Across two graduating years (2024/25 and 2025/26) — more than 40 projects.

- Help with problem framing, choice of architecture and quality metrics for the task type and class imbalance.
- Advise on the pipeline: data → EDA → features → model → service.
- Diagnose common mistakes: data leakage, invalid validation schemes, class imbalance, overfitting.
- Prepare projects for defence and run code reviews.

Domains: LLM/RAG and multi-agent systems, computer vision (MRI, X-ray, industrial inspection), time series and predictive maintenance, GNN and recommender systems, NLP.

Skills

LLM / NLP: LLM · RAG · LangChain · LangGraph · Hugging Face Transformers · vector search (ChromaDB, FAISS) · embeddings · prompt engineering · prompt-injection defence · LoRA / QLoRA · local models (Ollama)

ML / Data Science: Python · SQL (window functions) · pandas · NumPy · scikit-learn · gradient boosting (XGBoost, LightGBM, CatBoost) · stacking and blending · feature engineering (tsfresh, TSFEL) · SHAP · applied statistics and hypothesis testing · A/B design · EDA and visualisation · time series · anomaly detection · leakage-free cross-validation

Deep Learning: PyTorch · TensorFlow / Keras · CNN · RNN / LSTM / GRU · U-Net · transfer learning and fine-tuning (VGG16, Xception)

Backend / MLOps: FastAPI (async) · Flask · Django REST Framework · REST API design · Docker and docker-compose · PostgreSQL · Git · Linux · pytest · p50/p95/p99 latency metrics

Engineering practice: reproducible experiments · honest baselines · task-appropriate metric selection · code review · technical documentation

Education

Master's — software engineering, track «Artificial Intelligence in Digital Product Development». ITIS KFU, Kazan, 09.2025 — 06.2027.

Bachelor's — software engineering, track «Information Systems Development Technologies». ITIS KFU, Kazan, 09.2021 — 06.2025. Graduated with honours, GPA 4.94/5.0.

Professional retraining «Fundamentals of Data Platform Development», KFU, 252 hours, 2025.

Certificates

- Prize winner, Open Doors: Russian Scholarship Project international olympiad, track «Applied Mathematics and Artificial Intelligence», 2025
- Best international graduate, ITIS KFU, 2025
- All-Russian Mathematics Dictation, 15/15 (T-Education), 2025
- Semifinalist, Volga-IT'24 olympiad, «AI and Data Analysis» track, 2024
- Speaker Certificate, Sharjah International Conference on AI & Linguistics (UAE), 2024
- «Digital Curator», KFU, 2024

Languages

Arabic — native · English — C2 · Russian — C1